

# **Sri Lanka Institute Of Information Technology**



**Professional Skills**

**IT2090**

**Personal Development Plan**  
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**Bsc (Hons) Information Technology**  
**Year 2 Semester 2**

## 1. Self – Assessment:

I have done a self-assessment via SWOT analysis and skills audit to look at where I am today and identify areas of improvement in the domain of data science.

### Strengths:

I have built a solid foundation in the fundamentals of data science including statistics, machine learning principles and data visualization. I'm becoming more proficient with real-world datasets through disambiguation of GitHub tools so I could explore methods of analysis. I am learning new frameworks and tools quickly: visualization tools such as Matplotlib and Seaborn, and tools and libraries that are integral features of Python, including Pandas, NumPy, and clustered data through Scikit-learn. Furthermore, my strong analytical and critical reasoning skills are assisting me in successful completion of complex data driven problems. Moreover, the certifications I have successfully passed are allowing me to build credibility and value onto my technical resume.

### Weaknesses:

I experience anxiety when speaking to large groups of people, and I struggle to communicate my results effectively. I also have not developed competent time management skills and cannot multitask across several projects. An area of development is incorporating coding skills in R, SQL, and Python (Advanced features) as I use them for data analysis or automation tasks. I have only one internship to accompany my academic projects, which signifies that I still require industry-level practical experience.

### Opportunities:

I have a variety of opportunities to learn through university workshops, data science clubs, internships, hackathons, or Kaggle competitions. I get to put my learning in practice, gain "real world" experience, and build my portfolio from these opportunities. If I also have advanced certifications in machine learning, deep learning, and cloud data platforms (AWS, Azure, Google Cloud) then I will be more marketable.

### Threats:

Data science is an extremely competitive industry and there are large number of educated graduates who will be competing for the same job opportunities. The rapid advances in technology being made in AI, machine learning and automation will constantly require upgradability. Balancing study, skills development, and career activities would not always be easy or possible.

As I have used the SMART structure I have proposed below, short-, medium- and long-term goals that are Specific, Measurable, Achievable, Relevant and Timebound.

#### Short-term (6–12 months):

- Commit to increasing parity speaking engagement to 2 speaking engagements.
- Achieve a certification. (ex: google Data Analytics).

#### Medium-term (1–3 years):

- Obtain relevant work experience within an internship in data science by 2027.
- Sponsor a university data project to demonstrate leadership.

#### Long-term (3–5 years):

- Obtain an advanced certification (e.g., TensorFlow, AWS ML). Action Planning

## **2. Goal Setting (SMART Goals):**

I have planned a practical way to work towards my goals in data science. I will work on my public speaking skills by getting involved in the university debate club or getting involved in a data science club and with workshops and practicing with recorded public speaking. After working on my public speaking skills, I will focus on the certification aspect and finish Google Data Analytics or Power BI in mid-2026 where I will spend a couple of hours a week studying. I will also work on my hands-on skills through a minimum of three Kaggle competitions and/or hackathons and put them in a portfolio. As I work on my hands-on skills, I will build my employability by updating my LinkedIn and CV and applying to data science internships at the beginning of 2024. I will also use opportunities to lead a project group in the university to bolster my leadership/working in a team skills. Lastly, I will make sure I have regular time each week to pair formal study and academic experience of skills with hands-on projects which will support my coding skills in Python, SQL and R and support my employability with projects that are useful, for example, building data cleaning tools, dashboards and predictive models.

## **2. Monitoring & Evaluation Strategy**

I will keep a reflective journal once a month in which I will note milestones, challenges that I faced, and where I could improve. I will also use a structured tracker on Notion or Excel which I will populate with milestones (i.e., certificates, projects, and Kaggle competitions) and when they were completed in order to track my progress. I will also engage in consistent feedback from my lecturers, mentors, and peers - especially after my presentations and group projects - to be able to support the progression of my skills with feedback from an external party. Lastly, I will schedule quarterly reviews of my goals and if any one goal falls behind by more than two months, I will be able to evaluate my approach, amend timelines, and use other strategies to make sure I am on track.

## **3. Personal Insight & Motivation**

My passion for data science is based upon a sincere curiosity always exists in my mind, how can data be altered into translatable visions and techniques that can effectively address real world problems. I enjoy the intellectual challenge of solving issues, recognizing patterns to create informed insights and using reasoning to optimize value. I appreciate doing hands-on activities like Kaggle challenges, GitHub projects, and projects and assignments at university, all of which have helped develop my curiosity and willingness to improve by employ my technical skills and signalling what I needed to develop. For instance, a predictive modelling project would assist in developing my Python skills, and keep my attention on machine learning whilst a data visualization project would help me develop skills around clear communication and a narrative approach to data. Overall, all of these experiences helped me develop connections through theory to practice and continue to keep me engaged and wanting to learn.

I appreciate the opportunity to work as part of a team and independently as it relates to data science for proposing an experimental plan with algorithms, for hypothesis testing, and for a better operational technical understanding of the work I will be doing. In each of my group projects, I was able to appreciate the perspectives of others, talk about multiple things at once, and not only think about frameworks for leadership and communication, but also my own development as an expert with technical skills.

In retirement, I can see myself growing into an able and confident data scientist/machine learning engineer, working on relevant, meaningful work in healthcare, finance or technology. While I will be leveraging my skills to help businesses get farther ahead, stored in written business problems to solve, I still want to tackle societal problems that deal with issues such as

improving health care forecasting, improving productivity of resources, and improving sustainability. I look at this as an opportunity to keep learning, keep up with technology, and develop my know how so I can help advance the profession as I move into a leadership position.

#### **4. Presentation & Communication**

The Personal Development Plan (PDP) has clear headings and organized sections to optimize readability and logical flow. Bullet points and tables were used wherever applicable to concisely present information as a summary, making them easier to read. In addition, the PDP will include a Gantt chart presentation to visually demonstrate the timeline for certifications, internships and skill-building activities to track development progress more efficiently.

Overall, the PDP presentation is professional, formatted and proofread that does not contain grammatically erroneous. This accomplishes not only communicating ideas clearly; but showing due diligence and communication skills on how the PDP meets professional and academic standards of quality with attention to detail that is relevant to Improving Communication Skills through Personal Development Plan.

#### **5. Use of Tools & Evidence (Optional)**

To verify and support the aims outlined in this PDP, I will attach relevant tools and evidence as documentation. Documentation will include screenshots of Kaggle competitions, GitHub projects and other hands-on experiences to illustrate my participation in data science activities. Documentation to support activity, like certificates received for completed courses and exams, like those from Google regarding Data Analytics or Power BI will be included as an appendix to authenticate my accomplishments and demonstrate my increasing technical ability.

A link to my updated LinkedIn profile will be included to demonstrate continued professional development and evidence of my involvement with professional networks and project contributions. Furthermore, a copy of my draft CV will be attached to represent my academic credentials, technical capabilities and hands-on experience, to ensure all documentation is relevant to my career goals as a data scientist.

#### **6. Learning and Development Strategy**

In pursuit of my professional development I will focus on structured pathways towards development that includes online learning, real projects, and some level of immersion into the industry. Platforms such as Coursera, Udemy, Kaggle, and others will provide a framework for mildly structuring professional development through developing skills in my identified topics of interest such as: Python programming as a general language, machine learning, data analysis, or data visualisation and presentation. By undertaking real projects or real world challenges I can apply theoretical knowledge to real world problems, improve my problem solving skills, and build a portfolio of work that demonstrates my technical ability. I will continue to seek out workshops, boot-camps, and events regarding Data Science which will provide an educator with the opportunity learn about what I am passionate about and where I can help, as well as help me get practical experience and meet others who are part of the community in which I will be teaching.

I also find it very interesting to learn about the new areas in Data Science, specifically algorithmic thought and development, artificial intelligence, deep learning, cloud-based analytics, and big data technology. I am a lifelong learner and can make learning into a habit. As the field of data science is rapidly changing, being aware of the new tools, methodologies, applications, uses, etc. that can deliver solutions to current problems posed by educational

data implies a need to engage with the field and develop connections that can aid my search for professional development in the future, and might help me in a job search. I know that a combination of self-paced learning, practical experience, and engagement with the Data Science community is the best path to build my area of focus, which will help me adapt to current challenges and also prepare me to take on and look for future opportunities that may arise in the Data Science field as well as gain confidence that the project I am working on could grow in complexity or in some cases even get to be more complicated where I would have taken a leadership role.